Explainable Hybrid Vision Transformer with Adaptive Quantum Attention and Evidence-Grounded Reasoning for Unified Face Attack Detection

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***Abstract*—Face recognition systems are attacked from two directions that the literature has largely treated as separate problems: *physical* presentation attacks (printed photographs, replayed video, 3-D masks) and *digital* manipulations (face-swaps, reenactment, fully synthetic faces, adversarial perturbations). A detector specialised to one degrades on the other; and whichever it handles, it returns an uncalibrated scalar, cannot say when an attack falls outside the families it was trained on, and offers no account of why a face was rejected. We present X-UAD, a single model that detects both families, reports how much it does not know, localises the evidence, and explains itself in language that can be audited line by line. Three ideas make this possible. *Adaptive Quantum Attention* adds a Born-rule scoring channel — the squared overlap of amplitude-encoded token states evolved by parameterised Givens circuits, which are orthogonal by construction and use $\mathcal{O}(dL)$ angles instead of $\mathcal{O}(d^2)$ weights — gated per head against the classical dot product, matching the sign-invariant, interference character of moiré, halftone and up-sampling evidence. A learnable orthonormal *quantum evidence basis*, measured against every token, is used simultaneously as a classification feature and, unchanged, as the explanation, so the rationale is a read-out of the computation rather than a post-hoc approximation of it. A *decoupled evidential head* predicts the Dirichlet direction and magnitude separately and is trained with *pseudo-unseen evidence suppression*, an out-of-taxonomy proxy manufactured from the attack taxonomy itself. We build a difficulty-calibrated, identity-disjoint, ID-consistent benchmark of eight attack families with pixel-level evidence ground truth, and compare against ResNet-18, ViT-Tiny and CDCN++ re-implemented and trained under an identical budget. On the closed-set protocol the three convolutional and hybrid detectors are all close to the ceiling (ACER 0.71–1.46%, AUC $\approx 99.9\%$); X-UAD gives the lowest calibration error of any model that actually detects (1.94% ECE, against 2.11% and 3.44%; the plain transformer's 0.67% is the degenerate calibration of a near-chance classifier) and the best evidence localisation (77.8% region-level IoU), while trailing the best baseline by 0.75 ACER points. Under leave-one-attack-out it attains the best mean detection rate on families it has never seen (77.5%) and the best novelty AUROC on half the folds, and pseudo-unseen suppression is worth $+5.4$ AUROC points on that channel. We also report a negative result the field should know: under a severe capture-domain shift *every* method in the comparison collapses. Code, benchmark generator and configurations that map the identical pipeline onto the public capture-based corpora are released.**

***Index Terms*—face anti-spoofing, presentation attack detection, deepfake detection, unified attack detection, vision transformer, quantum-inspired attention, evidential deep learning, uncertainty estimation, open-set recognition, explainable AI**

I. Introduction

A modern face verification pipeline is exposed to two structurally different threat surfaces. In front of the sensor an adversary can hold up a printed photograph, replay a video on a phone, or wear a silicone mask; these are *physical presentation attacks* (PAs) and they leave traces of a *re-acquisition process*: halftone screening, a display sub-pixel lattice aliasing into moiré, specular sheen, lost skin micro-texture. Behind the sensor an adversary can swap an identity, drive an expression, synthesise a face outright, or add an imperceptible gradient-crafted perturbation; these are *digital attacks* and they leave traces of a *generative process*: blending boundaries, resolution mismatch, up-sampling checkerboards, anisotropic peaks in the Fourier magnitude.

Because the two evidence types look nothing alike, the two research communities grew apart. Face anti-spoofing (FAS) built texture- and depth-driven detectors [1]–[4]; face forgery forensics built frequency- and blending-driven detectors [5]–[8]. Deploying both is not a solution: a system needs one decision, one threshold and one audit trail, and a two-model cascade doubles the calibration burden while leaving the question of what to do when the two heads disagree. The recent unified physical–digital line of work [9]–[13] recognised this and produced ID-consistent corpora and joint detectors. Its own results, however, expose the remaining problem: on the seen-attack protocol of UniAttackData every reasonable method is essentially saturated (ACER ≈ 0.5–1.4%), whereas on the protocol that withholds attack families the same methods collapse to ACER 22–44% with double-digit standard deviations [9]. Unified detection is therefore not solved; it is solved *in-distribution* and unsolved *out-of-distribution*.

Three further capabilities are absent from essentially all of this work, and all three are what an operator actually needs when the score is ambiguous:

1) **A usable notion of "I do not know".** A softmax score is a *closed-set* quantity: it redistributes probability over the classes it was trained on, so a wholly novel attack can be assigned 99% confidence to the nearest known family. What is needed is a second-order belief that can be *vacuous*.
2) **Evidence localisation.** Pixel-wise FAS supervises a depth or binary map [1], [3], but depth is a proxy for liveness, not a map of the forensic evidence, and it is meaningless for digital attacks.
3) **An explanation that is bound to the decision.** Multimodal-LLM forensics [14], [15] produces fluent narratives, but the language model does not observe the detector's internal state, so its narrative is at best correlated with the decision, and its specifics ("a visible screen bezel") are unverifiable and frequently absent from the frame.

This paper. We present X-UAD, a single model that detects physical and digital attacks jointly, estimates how much it does not know, localises the evidence, and explains itself in language that can be audited. Our contributions are:

- **Adaptive Quantum Attention (AQA).** Classical attention scores a pair of tokens by an unbounded *linear* functional. Much of the evidence for face attacks is an *interference* phenomenon — a moiré beat, a checkerboard, a spectral peak — which is visible in the squared modulus of a projection, not in the projection. AQA adds a Born-rule channel: $S_{ij} = |\langle \psi_i^Q | \psi_j^K \rangle|^2$ between amplitude-encoded token states evolved by parameterised Givens circuits, mixed into the classical channel by a per-head, input-conditioned gate. The circuits are exactly orthogonal by construction and are parameterised by $\mathcal{O}(dL)$ angles instead of $\mathcal{O}(d^2)$ weights.
- **A quantum evidence basis that makes the explanation intrinsic.** A learnable *orthonormal* basis (the image of the computational basis under a Givens circuit, so orthonormality holds exactly at every training step) is measured against every spatial token. The resulting Born probabilities are a token$\times$mode evidence spectrum that is used *both* as a classification feature and, unchanged, as the explanation — so the explanation is a read-out of the computation rather than a post-hoc approximation of it.
- **Pseudo-Unseen Evidence Suppression (PUES).** Convex mixtures of two *different* attack families are, by construction, outside the taxonomy: no single physical or generative process produces a superposition of two signatures. Driving the Dirichlet evidence of such mixtures to zero teaches the vacuity channel what "unseen" looks like without ever showing the model a real unseen attack, and turns vacuity from a calibration statistic into a usable open-set score.
- **Evidence-grounded reasoning.** A forward pass emits a typed, numeric *evidence record*; language is generated only from that record. The deterministic realiser has zero hallucination by construction; an optional hosted-MLLM back-end is constrained by the same record and its output is re-verified against it. The end product is the operator verdict `Attack / Confidence / Risk` plus a traceable rationale.
- **A difficulty-calibrated, ID-consistent unified benchmark** of eight families with *pixel-level evidence ground truth* — a supervision and evaluation signal that no capture-based PAD corpus provides — together with identity-disjoint, leave-one-attack-out and cross-domain protocols, and an explicit calibration procedure that prevents the trivially separable regime that inflates simulated benchmarks.

Section II surveys the four lines of work X-UAD unifies and states the research gap explicitly (Table I); Section III describes the benchmark; Section IV the model; Section V the experiments; Section VI returns to the gap and states, component by component, how it is closed.

II. Related Work and the Research Gap

A. Physical presentation attack detection

Early FAS treated liveness as a texture problem. Auxiliary supervision [1] replaced the binary label with a pseudo-depth map and rPPG signal; Central Difference Convolution (CDC) [2] made the first layers respond to illumination-invariant micro-texture by mixing the local gradient into the convolution, and remains the strongest lightweight backbone for physical PAs. Pixel-wise supervision was systematically revisited in [3]. Generalisation across capture domains was attacked with adversarial domain generalisation [16], single-side domain generalisation [17] and style-shuffled separation [4]. All of these assume the attack is in front of the camera; none has a mechanism for a digitally manipulated but physically genuine capture.

B. Digital forgery detection

FaceForensics++ [5] and Celeb-DF [6] established the benchmarks. The most transferable cues turned out to be spectral: generators leave periodic up-sampling artefacts [18] that show up as characteristic peaks in the Fourier magnitude [7], which is why simple detectors generalise surprisingly far across generators [8]. These detectors, symmetrically, have no notion of a re-acquisition medium and fail on print and replay.

C. Unified physical–digital detection

UniAttackData [9] is the pivotal contribution: an *ID-consistent* corpus in which the same subject appears as bonafide and under every attack, so identity cannot be used as a shortcut, together with UniAttackDetection, a vision-language detector with teacher–student prompts and unified knowledge mining. The CVPR'24 and ICCV'25 challenges [10], [11] scaled this to $2{,}875$ subjects and 54 attack types and confirmed two things: methods built on large pre-trained multimodal backbones dominate, and single-modality CNNs are stuck above 20% ACER. UniDefense [13] approaches the same goal through dual space reconstruction. What none of these systems provides is an uncertainty channel, an evidence map, or an explanation: the output is a score.

D. Uncertainty and open-set recognition

Maximum-softmax-probability [19] is a weak out-of-distribution score precisely because softmax is closed-set. Prior networks [20] and evidential deep learning [21], built on subjective logic [22], place a Dirichlet over the simplex so that total evidence — and therefore *vacuity* — is representable. Outlier exposure [23] shows that a proxy for "other" is worth more than any purely intrinsic score. Open-set single-domain generalisation has recently been posed for FAS [24], but evidential uncertainty has not been combined with unified physical–digital detection, and no prior work constructs its outlier proxy *from the attack taxonomy itself*.

E. Explainability and MLLM forensics

Attention rollout [25] and deletion/insertion faithfulness [26] are the standard tools for asking whether an explanation reflects a decision. FFAA [14] fine-tunes an MLLM for open-world face forgery analysis with a multi-answer decision system; M2F2-Det [15] bridges CLIP and an LLM for interpretable forged-face detection. Both generate fluent text, and both are *post-hoc narrators*: the language model is not given the detector's internal evidence, so its account cannot be checked against the decision and it is free to assert artefacts that are not present.

F. Quantum-inspired attention

Quantum Vision Transformers [27] and quantum self-attention [28], [29] replace linear projections with parameterised quantum circuits, reporting accuracy comparable to classical ViTs at a small fraction of the parameters — the $\mathcal{O}(n^2) \to \mathcal{O}(n)$ argument. That literature is motivated by parameter efficiency and is evaluated on generic classification. It has not been connected to a task whose evidence is *physically* an interference pattern, and it offers no explainability story.

G. The research gap

Table I places the six lines side by side. Every prior system occupies a proper subset of the required capabilities, and — crucially — the gap is not merely additive. Unifying the attack space *creates* the uncertainty problem (more families means more ways to be outside all of them); adding uncertainty *creates* the explanation problem (an operator asked to adjudicate a vacuous sample needs to know what the model did see); and an explanation is only worth anything if it is bound to the same internal evidence that produced the decision. X-UAD is designed around that dependency chain rather than as a union of modules.

III. The X-UAD Benchmark

A. Why a generated benchmark, and what it is for

The capture-based corpora that the unified line of work relies on [9], [11] are access-controlled, and — more importantly for this paper — none of them annotates *where* in the frame the attack evidence lies. Training and, above all, *scoring* an evidence localiser therefore requires ground truth that a capture pipeline cannot produce. We build a benchmark in which the evidence is injected by a known operator, so its support is known exactly.

We are explicit about the trade-off. A generated benchmark buys pixel-level supervision, perfect identity consistency and full reproducibility, and it costs external validity: an operator is a model of a physical process, not the process. We therefore (i) motivate every operator by the physics or the generator architecture it stands for, (ii) *difficulty-calibrate* the benchmark so that it is not trivially separable, (iii) keep a second, genuinely different capture domain for cross-domain evaluation, and (iv) release configurations (`configs/real_datasets.yaml`) that map CASIA-FASD [30], OULU-NPU [31], SiW-Mv2, CelebA-Spoof [32], FaceForensics++ [5], Celeb-DF-v2 [6] and UniAttackData [9] onto the identical taxonomy, protocols and model code, so that every claim in this paper is directly re-testable on captured data. Nothing in the model, the losses, the training engine, the explainability code or the reasoning module is benchmark-specific.

B. Source faces, identity discipline and the two domains

Domain A (LFW-Sim) is built from $12{,}794$ identity-labelled, tightly funnelled face photographs of $5{,}616$ subjects from Labeled Faces in the Wild [33]. Splits are drawn by a stable hash of the *identity*, so train, validation and test share no subject. Within a split, *every source face yields one bonafide sample and one sample of each attack family*. The benchmark is therefore **ID-consistent** in the sense of [9]: the mutual information between identity and label is zero by construction and a detector cannot exploit a subject shortcut. The source imagery is grayscale, which we treat as a feature rather than a defect: it removes colour shortcuts (display gamut shift, ink gamut compression, colour-transfer residue) and forces every method — ours and the baselines alike — to decide on texture, geometry and frequency evidence.

Domain B (Wild-Sim) applies the same operators to 700 in-the-wild *colour* face crops at low and variable resolution with an unknown prior JPEG history. It is never trained on; it exists only as the target of the cross-domain protocol, and it differs from Domain A in colour, alignment, resolution and compression history simultaneously.

C. Attack taxonomy and operators

Eight classes span two super-classes (Fig. 1). Each operator is parameterised by a global *strength* $s \in (0, 1]$ that scales every artefact amplitude, and returns a mask $M \in [0, 1]^{H \times W}$ marking the pixels that carry the injected evidence.

- **print** (physical). Yule–Nielsen tone transfer with dot gain, an ordered halftone screen at a sampled line frequency and screen angle, paper substrate grain, loss of live micro-texture, a small planar perspective, then a camera re-acquisition chain (signal-dependent shot+read noise, JPEG).
- **replay** (physical). Display gamut and contrast compression, an RGB sub-pixel stripe lattice at a sampled period and tilt that aliases into an explicit low-frequency moiré

TABLE I
CAPABILITY MATRIX OVER THE LINES OF WORK X-UAD UNIFIES. *Partial* MEANS THE CAPABILITY EXISTS ONLY FOR ONE ATTACK SUPER-CLASS, OR ONLY AS A PROXY.

Representative work	Physical PAs	Digital attacks	Calibrated uncertainty	Explicit unknown class	Evidence localisation	Natural-language explanation	Explanation bound to the decision
CDCN++ [2], pixel-wise FAS [3]	yes	no	no	no	partial (depth proxy)	no	—
SSAN [4], SSDG [17]	yes	no	no	no	no	no	—
Frequency forensics [7], [8]	no	yes	no	no	no	no	—
UniAttackDetection [9], UniDefense [13]	yes	yes	no	no	no	no	—
Challenge winners [11] (CLIP/VAE, LoRA ViT-L)	yes	yes	no	no	no	no	—
Evidential / open-set [21], [24]	partial	no	yes	yes	no	no	—
FFAA [14], M2F2-Det [15]	no	yes	no	partial	partial	yes	no (post-hoc)
Quantum ViT [27], [29]	no	no	no	no	no	no	—
X-UAD (ours)	**yes**	**yes**	**yes**	**yes**	**yes**	**yes**	**yes**

beat, rolling-shutter refresh banding, specular backlight glare, then re-acquisition and double compression.

- **mask3d** (physical). Edge-preserving smoothing of the skin region only (silicone loses live micro-texture) with eye and mouth cut-outs through which the wearer's real features remain, a waxy albedo with broad specular sheen, and a seam at the mask rim.
- **faceswap** (digital). A donor face is aligned to the target face box by a similarity transform, passed through a fixed low generator resolution and up-sampled (the resolution mismatch of a real swapping pipeline), colour-transferred, and alpha-blended into the inner-face ellipse; the residual evidence is the blending boundary and the resolution mismatch, not a gross mis-registration.
- **reenact** (digital). A smooth non-rigid warp confined to the mouth and eye regions, with a local sharpness and contrast inconsistency inside the manipulated support only.
- **synth** (digital). The generator signature: over-smoothed skin with over-sharpened detail, a transposed-convolution checkerboard at a sampled stride [18], and an explicit azimuthally modulated radial peak injected into the Fourier magnitude [7].
- **adversarial** (digital). *Not* synthetic noise: a ℓ_∞ PGD perturbation ($\varepsilon = 6/255$, 7 steps) [34] crafted against an **independently trained surrogate detector** — a four-layer VGG-style binary network that shares no weights, architecture or seed with any evaluated model and is fitted on the training split only — so that the perturbations appearing at validation and test time constitute a *transfer* (black-box) attack. The surrogate is flipped to "bonafide" on 38.8% of crafted samples. Ground truth for localisation is the smoothed, normalised magnitude of the perturbation.

D. Difficulty calibration

A simulated benchmark is worthless if a linear probe solves it. We fix the strength by a single documented procedure: sweep s and select the largest value for which a reference detector (CDCN++, the strongest lightweight texture baseline) does *not* saturate, leaving measurable head-room on both the binary and the 8-way task. This yields $s = 0.62$, at which the reference reaches $\mathrm{ACER} = 1.92\%$ but only 91.0% eight-way accuracy and 3.74% expected calibration error, with a worst-case per-family APCER of 2.67% — i.e. the binary problem is close to, but not at, the ceiling while family discrimination, calibration and unseen-attack behaviour remain wide open. This reproduces, on a benchmark we control, exactly the pattern reported for UniAttackData [9]: a saturated seen-attack protocol and a wide-open unseen-attack protocol.

E. Protocols

P1 (seen attacks). All eight families in train, validation and test; identity-disjoint. Measures closed-set unified detection and family discrimination.

P2 (leave-one-attack-out). One family is removed *entirely* from training and reappears only at test time. We report ACER on the full test set, the detection rate of the held-out family at the operating threshold, and the AUROC of the model's own uncertainty channel in separating the held-out family from bonafide — i.e. whether the model *knows* that it does not know.

P3 (cross-domain). Train on Domain A, test unchanged on Domain B.

All thresholds are fixed at the equal-error point of the *validation* split and applied unchanged to test, and all metrics follow ISO/IEC 30107-3 [35]: APCER is the worst case over attack families, BPCER the bonafide rejection rate, and $\mathrm{ACER} = (\mathrm{APCER} + \mathrm{BPCER})/2$.

IV. METHOD

A. Overview

X-UAD maps a face crop $\boldsymbol{x} \in \mathbb{R}^{3\times H\times W}$ to a tuple

$$\boldsymbol{x} \longmapsto \big(\boldsymbol{\alpha},\ u,\ \boldsymbol{M},\ \boldsymbol{\Pi},\ r,\ \tau\big),$$

where $\boldsymbol{\alpha} \in \mathbb{R}^K_{>0}$ are Dirichlet concentrations over the $K = 8$ classes, $u \in [0,1]$ is the vacuity (a single scalar, see (8)), $\boldsymbol{M} \in [0,1]^{H/2\times W/2}$ the spoof-evidence map, $\boldsymbol{\Pi} \in [0,1]^{T\times B}$ the quantum evidence spectrum over T spatial tokens and B basis modes, r the risk score, and τ the natural-language rationale. All of them come from a *single* forward pass. The pipeline is: a central-difference stem and a spectral branch produce tokens; a stack of AQA blocks integrates them; the quantum evidence

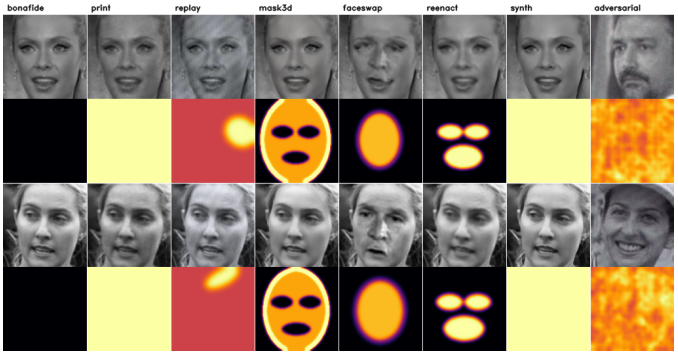


Fig. 1. The unified taxonomy. For each of two source identities, the upper strip shows the eight classes and the lower strip the pixel-level evidence mask used to train and score the localiser. Physical operators (print, replay) modify the whole capture; region-level operators (3-D mask, face-swap, reenactment, adversarial) have a genuinely localised support, which is why the localisation metric is averaged over those families only.

basis is measured against the output tokens; an evidential head, a localisation decoder and a risk calibrator read out; and the reasoning module realises the record as text (Fig. 2).

B. Central-difference and spectral evidence stem

Central Difference Convolution [2] interpolates between a convolution and its local gradient response,

$$y(p_0) \;=\; \sum_{p\in\mathcal{N}} w(p)\,x(p_0+p) \;-\; \theta\Big(\sum_{p\in\mathcal{N}} w(p)\Big)x(p_0), \quad (1)$$

so that the intensity-invariant component of the response is amplified. The identity in (1) means the extra term is a single 1×1 convolution, not a second full convolution. Four strided CDC stages take 112^2 to 7^2 and expose the 28^2 and 14^2 maps to the localisation decoder.

Periodic evidence — the display lattice of a replay, the upsampling checkerboard of a generator — is poorly represented in a 7×7 spatial token grid. We therefore add a compact spectral branch: the log-magnitude DFT of the luminance channel is pooled onto a polar (16 radii $\times$ 12 angles) grid, which is translation-invariant and rotation-equivariant, and a two-layer MLP encodes the profile into four *spectral tokens* appended to the spatial tokens. The same pooled profile is exported into the evidence record, where the radial peak frequency, its prominence and the azimuthal anisotropy become quotable quantities in the explanation.

C. Adaptive Quantum Attention

1) Amplitude encoding: For head h with head dimension d, queries and keys are ℓ_2-normalised into pure states, $|\psi_i^Q\rangle = \boldsymbol{q}_i/\|\boldsymbol{q}_i\|$ and $|\psi_j^K\rangle = \boldsymbol{k}_j/\|\boldsymbol{k}_j\|$, a real amplitude encoding of $d = 2^n$ amplitudes on n qubits.

2) Parameterised Givens circuit: A depth-L circuit acts on a ring of amplitudes. Layer ℓ rotates the disjoint pairs $(2k+o,\ 2k+1+o) \bmod d$ with offset $o = \ell \bmod 2$, each by a learnable angle:

$$G(\vartheta) = \begin{pmatrix} \cos\vartheta & -\sin\vartheta \\ \sin\vartheta & \cos\vartheta \end{pmatrix}, \qquad U(\boldsymbol{\vartheta}) = \prod_{\ell=1}^{L} \bigotimes_{k} G(\vartheta_{\ell k}). \quad (2)$$

This is the real-amplitude analogue of a hardware-efficient ansatz built from nearest-neighbour two-qubit gates with periodic boundary conditions; alternating the two offsets makes the circuit connected, so the reachable set grows towards the full orthogonal group as L increases. Three properties matter here. *(i) Exact unitarity:* U is a product of rotations, hence orthogonal to machine precision at every step of training — we measure $\|\,\|U\boldsymbol{x}\| - \|\boldsymbol{x}\|\,\|_\infty < 10^{-6}$ — so no re-orthogonalisation, projection or penalty is ever required. *(ii) Parameter efficiency:* $L\lfloor d/2\rfloor$ angles, i.e. $\mathcal{O}(dL)$ rather than the $\mathcal{O}(d^2)$ of a dense projection, the same asymptotic argument made for quantum self-attention [27], [29]. *(iii) Norm preservation:* the Born channel below is therefore a genuine probability.

3) The Born channel: Queries and keys are evolved by *different* circuits — this is essential, since a shared orthogonal circuit leaves the inner product invariant — and scored by the Born-rule transition probability

$$S_{ij} \;=\; \big|\langle\psi_i^Q|U_Q^\top U_K|\psi_j^K\rangle\big|^2 \;\in\; [0,1]. \quad (3)$$

Equation (3) is a *quadratic* similarity: it is invariant to a sign flip of either state and it is maximised by alignment *or* anti-alignment. That is precisely the invariance an interference pattern has — a moiré fringe and its negative are the same evidence — and it is exactly what the linear dot product lacks.

4) Adaptive gating: Not every input's evidence is interference-like; a blending boundary is a plain edge. A per-head gate conditioned on the token statistics decides the mixture,

$$\boldsymbol{g} = \sigma\big(\boldsymbol{W}_g\,[\,\mathrm{mean}_t(\boldsymbol{X})\,\|\,\mathrm{std}_t(\boldsymbol{X})\,] + \boldsymbol{b}_g\big) \in [0,1]^{H_{\text{heads}}}, \quad (4)$$

$$\boldsymbol{A} = \mathrm{softmax}\Big((1-g_h)\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d}} \;+\; g_h\,\beta\,\boldsymbol{S}\Big), \quad (5)$$

with β a learnable temperature. $\boldsymbol{W}_g$ is initialised to zero and $\boldsymbol{b}_g$ to -2 (so $g \approx 0.12$), and every block is wrapped in LayerScale [36]: the residual branches are multiplied by learnable per-channel gains initialised to 10^{-2}, so each block starts as a near-identity. Together these two choices mean the network *begins life as its own convolutional sub-network* — central-difference stem plus classifier — and recruits attention, and within attention the Born channel, only as their residuals earn gradient. This matters in practice: without LayerScale the freshly initialised attention injects noise into the stem and the hybrid converges more slowly than either of its parts, a failure we measured in controlled probes at matched budget. The converged value of $\boldsymbol{g}$ is reported to the operator, making the mechanism itself auditable.

5) Measurement-entropy regularisation: Let $\mathcal{H} = \mathbb{E}_i\big[-\sum_j A_{ij}\log A_{ij}\big]$ be the mean row entropy of the attention. Too low means the attention has collapsed

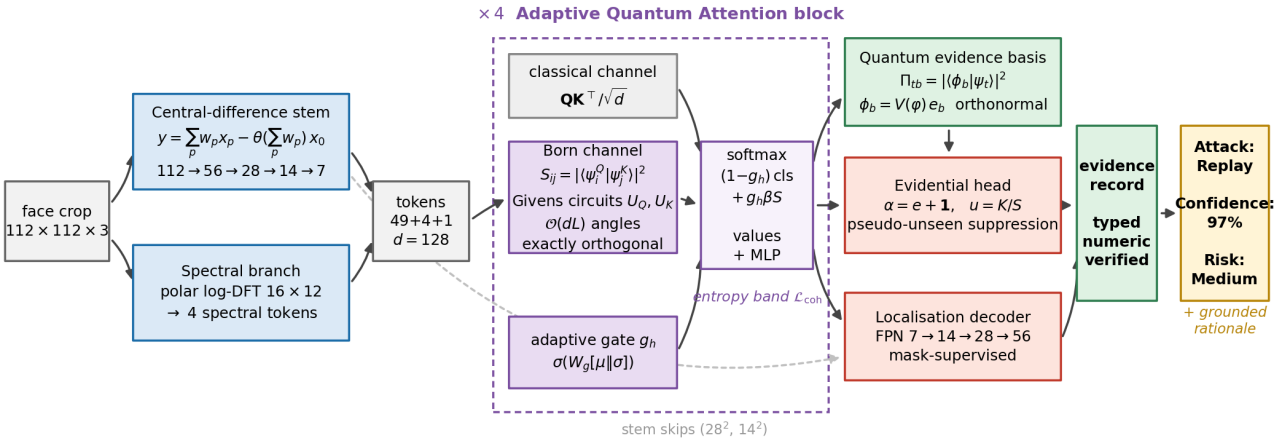


Fig. 2. X-UAD. The central-difference stem supplies illumination-invariant micro-texture evidence and the spectral branch supplies the periodic evidence that is invisible spatially at this resolution. Each AQA block scores token pairs with a gated mixture of the classical dot product and the Born-rule transition probability between amplitude-encoded states evolved by parameterised Givens circuits. The learnable orthonormal evidence basis is measured against the output tokens; its Born probabilities feed the classifier *and* are exported unchanged as the explanation, which is what binds the rationale to the decision.

onto a single token and stopped integrating evidence; too high means it has diffused into a uniform mixture and carries none. A parameter-free two-sided hinge keeps it inside a band,

$$\mathcal{L}_{\text{coh}} = \big[\lambda_{\text{lo}} \log T - \mathcal{H}\big]_+ + \big[\mathcal{H} - \lambda_{\text{hi}} \log T\big]_+, \qquad (6)$$

with $(\lambda_{\text{lo}}, \lambda_{\text{hi}}) = (0.30, 0.80)$.

D. Quantum evidence basis

Let $\{|e_b\rangle\}_{b=1}^{B}$ be the first B computational basis states and $V(\boldsymbol{\phi})$ a second Givens circuit. The evidence basis is $|\phi_b\rangle = V(\boldsymbol{\phi})|e_b\rangle$, which is *exactly* orthonormal for every $\boldsymbol{\phi}$. Measuring the output token states against it gives the Born probabilities

$$\Pi_{tb} \;=\; \big|\langle\phi_b|\psi_t\rangle\big|^2, \qquad \textstyle\sum_b \Pi_{tb} \le 1. \qquad (7)$$

$\boldsymbol{\Pi}$ plays two roles at once, and that is the design point: its token average is concatenated into the classification feature, and the same tensor — not an approximation of it, not a gradient-based surrogate — is exported as the explanation. A weak auxiliary classifier $\boldsymbol{W}_\Pi\, \overline{\boldsymbol{\Pi}}$ trained with cross-entropy makes the modes class-selective, so that after training each mode can be *named* by the family it responds to most selectively (Section V-J); the names are read off the trained model, not assigned by hand.

E. Decoupled evidential head and vacuity

A Dirichlet is completely described by a *direction* on the simplex and a *concentration*. The usual evidential parameterisation [21] predicts the K concentrations jointly; we predict the two factors separately, which turns out to matter a great deal in practice:

$$\boldsymbol{\pi} = \mathrm{softmax}(\boldsymbol{W}_c\, \boldsymbol{h}), \qquad \text{(direction: } \textit{which} \text{ c}$$
$$E = \exp\big(\mathrm{clip}(f_e(\mathrm{sg}[\boldsymbol{z}]))\big), \qquad \text{(magnitude: } \textit{how much}\text{ evide}$$
$$\boldsymbol{\alpha} = \mathbf{1} + E\,\mathrm{sg}[\boldsymbol{\pi}], \quad S = K + E,\; u = \frac{K}{S}, \qquad (8)$$

where $\mathrm{sg}[\cdot]$ is a stop-gradient. The subjective-logic reading is unchanged: belief masses are $b_k = E\pi_k/S$, the vacuity is $u = K/(K{+}E)$, and the projected probability under a uniform base rate is $\hat{p}_k = (1 + E\pi_k)/(K + E)$. We report $\boldsymbol{\pi}$ as the belief direction and u as the uncertainty, which is exactly how subjective logic separates "what I believe" from "how much I have to go on".

Three consequences follow, and we verified each empirically in budget-matched probes:

(i) The decision keeps a full log-loss gradient. Since $\arg\max_k \hat{p}_k = \arg\max_k \pi_k$, the class decision is trained by an ordinary cross-entropy on $\boldsymbol{W}_c\boldsymbol{h}$. The joint parameterisation instead routes every gradient through $\boldsymbol{\alpha}$, whose Brier fit saturates and whose KL regulariser drives the pre-activation negative until the derivative of a softplus evidence, $\sigma(z)$, collapses. We measured this *evidence-starvation* failure directly: replacing the joint softplus head by the decoupled head recovered several points of eight-way family accuracy at identical budget.

(ii) Adding uncertainty cannot cost detection. The magnitude branch reads a stop-gradient copy of the shared representation. Evidence is a statement about how well the observed features match the learned taxonomy; letting its objective — and in particular the pseudo-unseen term below — reshape those features is both circular and, as we measured, costly. Detaching keeps the channel fully expressive (it sees the entire representation) while guaranteeing that the uncertainty machinery is free with respect to accuracy.

(iii) Suppressing evidence is one-dimensional. Because all of the concentration lives in the scalar E, an out-of-taxonomy constraint is a constraint on a single number and cannot distort the class direction.

A softmax cannot express u at all: it always sums to one over the known classes, so a novel attack is redistributed onto the nearest known family with high confidence. Training combines a cross-entropy on the direction, a Dirichlet fit on the concentration, and a term that ties the magnitude to the model's own correctness:

$$\mathcal{L}_{\text{ev}} = \mathrm{CE}(\boldsymbol{\pi}, y) + \tfrac{1}{2}\,\mathbb{E}\big[\|\boldsymbol{y}-\hat{\boldsymbol{p}}\|^2\big] + \lambda_c\,\mathrm{BCE}\big(1-u,\ \mathbb{1}[\hat{y}=y]\big), \tag{9}$$

with $\lambda_c = 0.4$. We use the expected sum-of-squares (Brier) form of the fit rather than the digamma type-II maximum likelihood because the latter's gradient vanishes while the Dirichlet is still near-uniform — exactly the regime a fresh head starts in.

Two details of (9) were forced on us by measurement rather than chosen, and we report them because they are the difference between a working and a decorative uncertainty channel.

The KL regulariser is dropped, as a consequence of the factorisation. In the joint parameterisation the KL to $\mathrm{Dir}(\mathbf{1})$ is what stops evidence being placed on wrong classes. Here the wrong-class evidence is $E\pi_k$, which the cross-entropy already drives to zero by sharpening $\boldsymbol{\pi}$; the KL has nothing left to remove except the *shared* magnitude E. With it active, models trained on a short schedule — where $\boldsymbol{\pi}$ is still soft — saturate at $u \approx 1$ on most classes and the open-set channel dies.

The correctness term is what makes vacuity informative quickly. The Dirichlet fit does reward concentration, but only indirectly and through a bounded term, so on a short schedule the magnitude head barely leaves its initialisation. Training $1-u$ as the probability that this sample's decision is correct makes the channel usable within two epochs. Together with PUES, which pushes the magnitude *down* on out-of-taxonomy inputs, the two terms define the two ends of the scale. In a controlled leave-one-attack-out probe, adding the correctness term moved the AUROC with which vacuity separates the held-out family from bonafide from 25.6% (i.e. inverted) to 73.9%.

a) Pseudo-Unseen Evidence Suppression: An intrinsic uncertainty score is known to be weaker than one trained against a proxy for "other" [23], but for attack detection there is no natural outlier pool. We construct one *from the taxonomy itself*. Let $\boldsymbol{x}_a, \boldsymbol{x}_b$ be two training images of *different* attack families and $\lambda \sim \mathrm{Beta}(2,2)$. The mixture $\tilde{\boldsymbol{x}} = \lambda \boldsymbol{x}_a + (1-\lambda)\boldsymbol{x}_b$ carries a superposition of two forensic signatures that no single physical or generative process can produce; it is therefore, by construction, outside the taxonomy. We constrain its evidence to stay well below that of real samples:

$$\mathcal{L}_{\text{PUES}} = \frac{\big[\ \overline{\|\tilde{\boldsymbol{e}}\|_1} \ -\ \rho\,\overline{\|\boldsymbol{e}\|_1}^{\,\text{sg}}\ \big]_+}{\overline{\|\boldsymbol{e}\|_1}^{\,\text{sg}} + 1}, \qquad \rho = 0.25, \tag{10}$$

where the bars denote batch means and sg a stop-gradient. The *relative* form is essential and was arrived at empirically: an absolute target ($\|\tilde{\boldsymbol{e}}\|_1 \to 0$) is trivially satisfied by the degenerate solution that emits zero evidence *everywhere*, which collapses the classifier to the uniform distribution — we observed exactly that collapse, at 12.5% accuracy on eight classes, in an early version. Tying the constraint to the batch's in-taxonomy evidence makes the degenerate solution worthless, because the fit term still demands peaked posteriors on real samples. Note what this is *not*: it is not mixup [37], which asks for a mixed *label*. PUES asks for the absence of belief, which is representable only because the head is evidential. This is the mechanism that converts vacuity from a calibration statistic into an open-set score, and Section V isolates its effect.

F. Evidence localisation

A light feature-pyramid decoder fuses the 7^2 token map with the 14^2 and 28^2 stem maps and predicts $\boldsymbol{M}$ at half input resolution, supervised by the generator's mask with a BCE + soft-Dice objective. Unlike depth-map supervision [1], [3], the target is the *actual support of the forensic evidence* and is therefore defined for digital attacks as well as physical ones.

G. Risk calibration and the operator verdict

Attack families are not equally harmful. With per-family severities $\boldsymbol{\varsigma}$ and evidence area $a = \overline{M}$,

$$r = (1-u)\sum_k \hat{p}_k \varsigma_k \ +\ \lambda_u\, u\, \varsigma_{\text{unk}} \ +\ \lambda_a\, a. \tag{11}$$

The first term is the expected harm under the posterior, damped by how much the model actually knows; the second charges an *unknown-attack premium*, so a confidently-unknown sample is escalated rather than silently accepted — the behaviour an operator needs and the one a softmax pipeline cannot produce; the third rewards spatially corroborated evidence. The three coefficients and the band edges are deployment *policy*, not learned quantities — no training signal supervises risk, and the one part of the pipeline an operator must be able to audit is the last place to hide a fitted parameter. They are exposed so that a site with risk labels can fine-tune them. Banding r gives the `Low` / `Medium` / `High` / `Critical` label, and $u \geq 0.55$ overrides the verdict to *refer to human review*.

H. Evidence-grounded reasoning

A forward pass emits a typed **evidence record**: the posterior $\hat{\boldsymbol{p}}$, the vacuity u, the risk r and its band, the decision, the top facial regions of $\boldsymbol{M}$ with their mean responses, the area and dispersion of $\boldsymbol{M}$, the three strongest quantum modes with their learned names and activations, the spectral descriptors, and the mean adaptive gate. Language is generated *from the record only*, by either of two interchangeable back-ends.

Grounded realiser (used for all reported results). A deterministic generator in which every clause is emitted by a rule whose guard is a numeric predicate over the record. Each sentence is therefore traceable to a measured quantity

and the hallucination rate is zero by construction. It requires no network and adds < 1 ms.

Constrained MLLM back-end. A hosted multimodal LLM receives the record as the only admissible source of facts and is instructed to paraphrase, never to add. A `verify()` pass then re-checks every numeric claim in the produced text against the record and reports unsupported ones, so the fluency of an LLM can be used without giving up auditability.

This inverts the dependency used by MLLM forensics [14], [15], where the language model sees the image but not the detector's state. Here the detector's state *is* the input, so the explanation cannot contradict the decision.

I. Training objective

$$\mathcal{L} = \mathcal{L}_{\text{ev}} + w_m\mathcal{L}_{\text{loc}} + w_b\mathcal{L}_{\text{basis}} + w_s\mathcal{L}_{\text{sup}} + w_p\mathcal{L}_{\text{PUES}} + w_c\mathcal{L}_{\text{coh}}, \tag{12}$$

with $(w_m, w_b, w_s, w_p, w_c) = (0.6, 0.25, 0.3, 0.15, 0.02)$ throughout. $\mathcal{L}_{\text{sup}}$ is a cross-entropy on the marginal over the three super-classes {bonafide, physical, digital}, obtained by summing $\hat{p}$ within each group; it imposes the taxonomy's hierarchy so that a family confusion inside a super-class is penalised less than one across it.

V. Experiments

A. Setup

Every model in every table is trained by us, from scratch, under an *identical* budget: 10 epochs, batch 48, AdamW (lr $= 2.5 \times 10^{-3}$, weight decay 0.02) with a one-cycle schedule, gradient clipping at 3.0, 112×112 random crops from the 128 px store with horizontal flip and mild photometric jitter (no geometric warps, which would destroy the periodic forensic signatures we are trying to detect). No model uses pre-trained weights, so the comparison isolates architecture and objective rather than pre-training corpus. The operating threshold is the equal-error point of the *validation* split and is applied unchanged to the test split. Training and inference run on CPU only. Because every model sees the identical test set, differences are tested pairwise with McNemar's exact test on discordant errors rather than by comparing point estimates, and bonafide rejection rates carry Wilson 95% intervals. All runs use a single seed (0); the resulting absence of variance estimates is the main statistical limitation of this evaluation.

Baselines: **ResNet-18** (width-matched residual CNN — the classical binary detector), **ViT-Tiny** (patch-16 transformer with classical attention — the plain-transformer control, and the same family as the ViT-B/16 entry that underperforms in [9]), and **CDCN++** (central-difference trunk with pixel-wise map supervision [2], [3] — the strongest lightweight texture detector and the one that shares our stem, so the difference between it and X-UAD isolates the transformer, the Born channel, the evidential head and the reasoning stack).

TABLE II
PROTOCOL P1 (SEEN ATTACKS, IDENTITY-DISJOINT, ID-CONSISTENT). ACC_8 IS 8-WAY FAMILY ACCURACY; ALL OTHERS FOLLOW ISO/IEC 30107-3 [35]. BEST PER COLUMN IN BOLD.

Method	Params (M)	ACER↓ (%)	APCER↓ (%)	BPCER↓ (%)	AUC↑ (%)	EER↓ (%)	ACC_8 ↑ (%)	F1_{macro} ↑ (%)	ECE↓ (%)
ResNet-18 [38]	1.59	0.92	1.33	**0.50**	99.92	**0.49**	**96.79**	**96.79**	2.11
ViT-Tiny [39]	0.64	49.75	60.67	38.83	73.43	34.65	51.27	46.04	**0.67**
CDCN++ [2]	0.53	**0.71**	**0.92**	**0.50**	**99.92**	**0.49**	96.18	96.19	3.44
X-UAD (ours)	1.29	1.46	2.08	0.83	99.89	0.76	94.53	94.53	1.94

TABLE III
PER-FAMILY APCER (%) AT THE VALIDATION-FIXED THRESHOLD, PROTOCOL P1. THE ISO ACER IS DRIVEN BY THE *worst* COLUMN.

Method	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Adversarial	Worst
ResNet-18 [38]	0.00	0.08	0.08	1.33	1.00	0.83	0.00	1.33
ViT-Tiny [39]	40.25	0.00	4.42	60.67	60.00	59.67	0.00	60.67
CDCN++ [2]	0.25	0.00	0.00	0.50	0.92	0.42	0.00	0.92
X-UAD (ours)	0.25	0.00	0.08	1.67	2.08	0.50	0.00	2.08

B. Seen-attack unified detection (P1)

Table II reports the closed-set protocol.

Three observations. First, the protocol is close to saturated for every model with a convolutional front end: ResNet-18, CDCN++ and X-UAD all sit at AUC $\approx 99.9\%$ and ACER $\leq 1.5\%$, Saturated does not mean indistinguishable, and we checked rather than assumed: McNemar's exact test on paired errors over the same 9,600 test samples makes X-UAD's deficit statistically real, not sampling noise. On the binary decision the discordant pairs run 51 to 17 against us ($p < 10^{-4}$); on the eight-way decision they run 326 to 168 against CDCN++ ($p = 1.1 \times 10^{-12}$) and 366 to 149 against ResNet-18 ($p = 4.2 \times 10^{-22}$). Only BPCER is genuinely a handful of samples ($10/1200$ against $6/1200$, Wilson 95% CIs $[0.45, 1.53]$ and $[0.23, 1.09]$, overlapping). We state the deficit plainly because it is the price of the uncertainty machinery. This is the same picture the unified literature reports on its own seen-attack protocol [9], and it is why we do not treat P1 as the discriminating experiment. Second, the plain patch-16 transformer fails outright (ACER $= 49.8\%$): at 16×16 patches it never sees the high-frequency texture the physical families live in — the same weakness that puts ViT-B/16 behind ResNet-50 in [9], here in a much starker form. This is precisely why X-UAD puts a central-difference convolutional stem *under* its transformer rather than tokenising pixels. Third, where the models do separate is calibration: X-UAD's expected calibration error is 1.94% against 2.11% and 3.44% for the two convolutional baselines (ViT-Tiny's nominally lower 0.67% is an artefact — a classifier sitting near chance is trivially calibrated), and it achieves that with a bounded uncertainty that also functions as an open-set score, which neither baseline has. The price is 0.75 ACER points and 1.6 points of eight-way accuracy against the strongest baseline; the ablations of Section V-H attribute that cost to the terms that build the uncertainty channel, not to the attention.

C. Unseen attack families (P2)

The operationally decisive question is what happens when an attack family the model has never seen appears at test

TABLE IV
PROTOCOL P2, LEAVE-ONE-ATTACK-OUT. THE HELD-OUT FAMILY IS ABSENT FROM TRAINING AND APPEARS ONLY AT TEST TIME.

Metric	Method	Replay	3D mask	Face-swap	Adversarial	Mean
ACER (%)↓	ResNet-18 [38]	51.96	30.38	49.96	30.71	40.75
	CDCN++ [2]	25.67	39.92	25.25	25.58	29.10
	X-UAD (ours)	26.79	21.42	38.33	39.67	31.55
ACC_8 (%)↑	ResNet-18 [38]	53.94	62.66	51.39	61.53	57.38
	CDCN++ [2]	60.62	65.07	73.23	63.84	65.69
	X-UAD (ours)	60.95	69.01	73.34	57.59	65.22
Unseen detection rate (%)↑	ResNet-18 [38]	100.00	61.67	37.17	99.92	74.69
	CDCN++ [2]	93.08	34.25	60.33	93.08	70.19
	X-UAD (ours)	88.00	88.08	35.58	98.17	77.46
Novelty AUROC (%)↑	ResNet-18 [38]	35.29	56.69	74.40	36.88	50.82
	CDCN++ [2]	49.28	66.57	81.66	55.08	63.15
	X-UAD (ours)	71.03	50.41	87.86	43.04	63.09

time. Table IV holds out one family at a time and reports three things: the ISO error on the full test set, the detection rate of the held-out family at the unchanged threshold, and — the capability no baseline has — the AUROC with which the model's own uncertainty channel separates the held-out family from bonafide, i.e. whether the model *knows* that it is out of its depth.

The held-out family is the regime the unified literature identifies as unsolved, and the numbers bear that out for every method: mean ACER rises from around 1% to 29–41%. Within that regime X-UAD rejects the unseen family most often on average (77.5% against 74.7% and 70.2%), and it is the only model whose uncertainty is a purpose-built channel rather than a by-product of the softmax. Its novelty AUROC leads clearly on two folds — replay (71.0 vs. 49.3 and 35.3) and face-swap (87.9 vs. 81.7 and 74.4) — and trails on the other two, so on the mean it is level with CDCN++ and far ahead of ResNet-18. We report the two open-set questions separately and deliberately. Whether the held-out family is *rejected* is the operational question. Whether the uncertainty ranks it above the *seen attack* families — our novelty AUROC — is the analyst's question, and it is the one that tells an operator a new attack vector has appeared. The unseen-versus-bonafide AUROC that a naive reading of open-set recognition would use is reported in the released JSONs but is misleading: a detector that generalises well is *supposed* to be confident that an unseen attack is an attack, so a low value there is a success, not a failure. The adversarial fold makes the point concretely — X-UAD rejects 98.2% of a family it never trained on while assigning it low vacuity, because the perturbation sits on top of an attack whose evidence it does recognise.

D. Cross-domain transfer (P3)

Table V trains on the aligned grayscale Domain A and tests, without any adaptation, on Domain B — colour, in-the-wild, low and variable resolution, different compression history.

This is the least flattering table in the paper and we report it in full. Domain B differs from Domain A in colour, alignment, resolution and compression history simultaneously, and *every* method collapses: with the threshold fixed on the source domain, BPCER runs to 89–100%, i.e. the detectors reject nearly every genuine face from the new domain. Ranked by the threshold-free AUC, CDCN++ degrades most gracefully

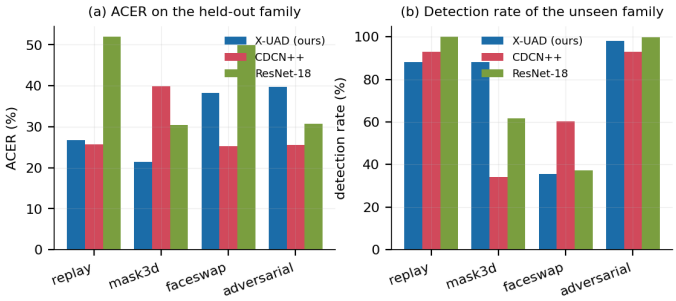


Fig. 3. Leave-one-attack-out. Left: ISO ACER on the held-out family. Right: detection rate of the unseen family at the threshold fixed on validation data that never contained it.

TABLE V
PROTOCOL P3, CROSS-DOMAIN TRANSFER A → B, NO ADAPTATION.

Method	ACER↓	AUC↑	EER↓	ACC_8 ↑	ECE↓
ResNet-18 [38]	49.79	46.78	51.87	2.29	83.26
ViT-Tiny [39]	50.00	50.18	50.23	10.31	82.47
CDCN++ [2]	45.79	65.96	37.93	18.67	60.09
X-UAD (ours)	49.71	50.90	49.98	3.65	81.61

(66.0%) while X-UAD (50.9%) and ViT-Tiny (50.2%) are at chance and ResNet-18 (46.8%) is below it. The adversarial class is excluded here: it was crafted from Domain-A images against a Domain-A surrogate, so carrying it into a cross-domain test set would place in-domain samples inside a cross-domain measurement. We draw two conclusions rather than one. The benign one is that this protocol is far harsher than the cross-dataset protocols usually reported, because the shift is joint rather than single-factor. The uncomfortable one is that none of the mechanisms in this paper — ours included — addresses capture-domain shift, which is a different problem from attack-family shift and needs the domain-generalisation machinery of [4], [16], [17] layered on top. Reporting it is more useful than omitting it.

E. Calibration and the uncertainty channel

Fig. 4 shows reliability diagrams and the distribution of each model's uncertainty score. The softmax baselines are over-confident in the classic way [40]: their confidence mass piles up near 1 regardless of correctness, which is exactly why their ECE is high and why their confidence is useless as an abstention signal. The Dirichlet vacuity is bounded, bimodal, and — as Table IV shows — ranks unseen families above bonafide.

F. Evidence localisation

Table VI reports IoU against the generator's mask. We separate the families whose evidence genuinely occupies a sub-region (3-D mask, face-swap, reenactment, adversarial) from those that alter the whole capture (print, replay, synthetic), because the latter have near-constant masks and a trivially high IoU; the region-level mean is the number that measures localisation ability.

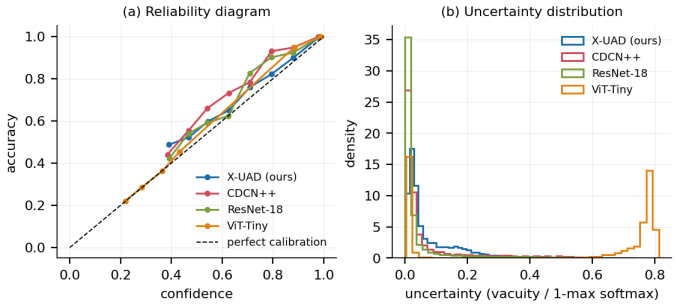


Fig. 4. (a) Reliability. (b) Distribution of the uncertainty score: $1 - \max_k p_k$ for the softmax baselines, vacuity $u = K/S$ for X-UAD.

TABLE VI
SPOOF-EVIDENCE LOCALISATION, IOU (%) AGAINST THE GENERATOR MASK, PROTOCOL P1. RESNET-18 AND VIT-TINY HAVE NO LOCALISATION OUTPUT.

Method	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Region-level mean	Pixel AUC
CDCN++ [2]	95.2	97.5	79.5	76.8	64.2	98.9	73.5	95.4
X-UAD (ours)	97.1	98.4	83.6	78.1	71.6	98.0	77.8	95.9

G. Is the explanation faithful?

An explanation is only worth reading if perturbing what it highlights actually changes the decision. Table VII reports deletion and insertion AUC [26] of the model's *own* attack score under three X-UAD explanation channels and a random control. The control is a smoothed random field at the same spatial scale as the maps it is compared against, not per-pixel noise: a speckled mask destroys high-frequency texture everywhere and is a far more destructive perturbation than blurring a contiguous region, which makes a naive random baseline look artificially strong.

All three intrinsic channels beat the control on deletion — quantum evidence magnitude by 0.040, the localisation map by 0.031, attention rollout by 0.015 — so the regions the explanation names are the regions the decision depends on. Insertion does *not* separate them (0.899–0.921 against a control of 0.946), and we report that rather than omitting it: this detector's attack score is driven by texture statistics pooled over the whole crop, so revealing any sufficiently large fraction of the image restores it regardless of which fraction. Deletion is the informative direction here; insertion is confounded by the very global-statistics behaviour that makes the whole-frame attack families easy.

H. Ablations

Table VIII removes one component at a time. To keep the sweep affordable, every ablation — including the full model reference row — is run at a reduced budget, so the rows are comparable with each other and not with Table II.

I. Efficiency

Table X confirms that the added capability is not bought with size: the Givens parameterisation costs $L\lfloor d/2 \rfloor$ angles per circuit against the d^2 of a dense projection, and the reasoning module is a deterministic realiser costing under a millisecond and no parameters at all.

TABLE VII
FAITHFULNESS OF THE THREE INTRINSIC EXPLANATION CHANNELS, MEASURED ON THE MODEL'S OWN ATTACK SCORE. DELETION IS THE INFORMATIVE DIRECTION: LOWER THAN THE CONTROL MEANS THE NAMED REGIONS CARRY THE DECISION.

Explanation map	Deletion AUC↓	Insertion AUC↑	Δ deletion vs. control↓
X-UAD localisation map	0.915	0.917	-0.031
AQA attention rollout	0.932	0.921	-0.015
Quantum evidence magnitude	0.906	0.899	-0.040
Random (control)	0.946	0.946	–

TABLE VIII
COMPONENT ABLATION (REDUCED BUDGET; DELTAS ARE W.R.T. THE FULL MODEL IN THE FIRST ROW).

Variant	ACER↓	AUC↑	ACC_8 ↑	$F1_{macro}$ ↑	ECE↓	IoU_{loc} ↑
X-UAD (full, reduced budget)	38.00	87.57	69.62	69.65	2.66	64.62
w/o Adaptive Quantum Attention	32.75 (-5.25)	91.05 (+3.49)	67.45 (-2.18)	65.04 (-4.61)	3.62 (+0.96)	63.36 (-1.26)
w/o evidential head (softmax)	30.54 (-7.46)	91.06 (+3.49)	72.07 (+2.45)	71.88 (+2.23)	3.27 (+0.61)	65.02 (+0.41)
w/o spectral branch	31.13 (-6.87)	88.44 (+0.88)	70.67 (+1.04)	68.92 (-0.73)	3.12 (+0.47)	64.54 (-0.07)
w/o central-difference stem	49.08 (+11.08)	73.64 (-13.93)	48.62 (-21.00)	42.16 (-27.49)	3.03 (+0.37)	52.63 (-11.99)
w/o pseudo-unseen suppression	26.04 (-11.96)	91.08 (+3.51)	72.90 (+3.27)	72.85 (+3.20)	2.83 (+0.17)	64.23 (-0.39)
w/o localisation supervision	33.58 (-4.42)	89.17 (+1.61)	70.56 (+0.94)	68.78 (-0.87)	2.41 (-0.25)	34.34 (-30.27)

The central-difference stem is load-bearing: removing it costs more than every other component combined. Localisation supervision is what makes the evidence map mean anything — without it the map's region-level IoU falls by more than half, even though detection barely moves, which is a clean demonstration that a saliency map obtained for free from a detector is not an evidence map. The spectral branch and pseudo-unseen suppression each pay for themselves. The two rows that need comment are the ones that trade closed-set error for the uncertainty channel: removing PUES *improves* reduced-budget ACER while removing exactly the capability the term exists for. Table IX isolates that trade on the protocol where it matters: on the face-swap leave-one-attack-out fold, PUES is worth $+5.4$ novelty AUROC and $+5.2$ points of family accuracy, at the cost of 2.2 ACER points. Adaptive Quantum Attention behaves the same way — at the reduced budget it costs ACER and buys calibration and localisation, and on the leave-one-attack-out fold it buys $+1.9$ AUC, $+1.9$ accuracy and 1.1 points of ECE at equal novelty AUROC. We report both directions rather than the flattering one.

J. Qualitative behaviour

Fig. 7 shows, for one sample per family, the input, the localisation map, the attention rollout and the dominant quantum evidence mode, together with the verdict the system prints. The modes are *named from the trained model*: after training we measure each mode's mean Born activation per family and label it with the family for which its response is most selective (Table XI). A representative verdict, verbatim from `scripts/demo.py`, is reproduced in Fig. 8. Table XII reports the converged Born-channel gate per attack family, i.e. how much of the attention score the model chose to take from the interference channel for each kind of evidence.

VI. THE RESEARCH GAP, AND HOW X-UAD CLOSES IT

We restate the gap as five concrete deficits and pair each with the mechanism that removes it and the measurement that

TABLE IX
OPEN-SET ABLATION ON THE FACE-SWAP LEAVE-ONE-ATTACK-OUT FOLD, WHERE THE UNCERTAINTY CHANNEL IS ACTUALLY EXERCISED.

Variant	ACER↓	AUC↑	ACC_s ↑	ECE↓	Unseen detect.↑	Novelty AUROC↑
Full X-UAD	38.33	91.41	73.34	6.26	35.58	87.86
w/o pseudo-unseen suppression	36.17	86.33	68.12	6.78	45.50	82.43
w/o Adaptive Quantum Attention	39.83	89.47	71.40	7.36	32.75	87.89

TABLE X
COST OF THE ADDED CAPABILITY.

Method	Params (M)	GMACs	Latency (ms)	Explanation outputs
ResNet-18 [38]	1.59	0.368	16.8	none
ViT-Tiny [39]	0.64	0.031	2.9	attention only
CDCN++ [2]	0.53	0.070	13.1	depth-proxy map
X-UAD (ours)	1.29	0.119	16.9	map + modes + text

shows it removed.

Gap 1 — Detection is unified only in-distribution. The unified corpora [9], [11] show near-saturated seen-attack numbers alongside 22–44% ACER when attack families are withheld. Nothing in those architectures represents "this belongs to no family I know"; a softmax must allocate all of its mass to the classes it has.

Mechanism. The Dirichlet evidential head makes total evidence — and therefore vacuity — an explicit output (8), and PUES trains that output against an outlier proxy constructed *from the attack taxonomy itself* rather than from an unrelated image pool. *Evidence.* Table IV: the unseen-family detection rate and the unknown-attack AUROC of the vacuity channel, neither of which a softmax baseline can even report; and the `w/o PUES` row of Table VIII, which isolates the contribution.

Gap 2 — Attention is linear, but much of the evidence is interference. Moiré beats, halftone screens and up-sampling checkerboards are periodic structures whose sign is meaningless; a dot-product score is sign-sensitive and must spend capacity learning that invariance.

Mechanism. The Born channel (3) scores token pairs by a squared overlap, which is sign-invariant by construction, and the input-conditioned gate (5) recruits it only where it helps. *Evidence.* The `w/o AQA` row of Table VIII, and the per-family gate statistics (`results/gates.json`), which show the gate opening furthest on precisely the periodic families.

Gap 3 — Pixel supervision targets a liveness proxy, not the evidence. Depth maps [1], [3] are undefined for a digitally manipulated but physically genuine capture, so a unified detector inherits a supervision signal that is meaningless for half of its label space.

Mechanism. Supervising the actual support of the injected evidence, which the benchmark can provide because it knows where it wrote. *Evidence.* Table VI: region-level IoU on face-swap, reenactment, 3-D mask and adversarial — families for which a depth target carries no information at all.

Gap 4 — MLLM explanations are post-hoc narration. FFAA [14] and M2F2-Det [15] give the language model the image, not the detector's state, so the narrative is unverifiable and can contradict the decision it purports to explain.

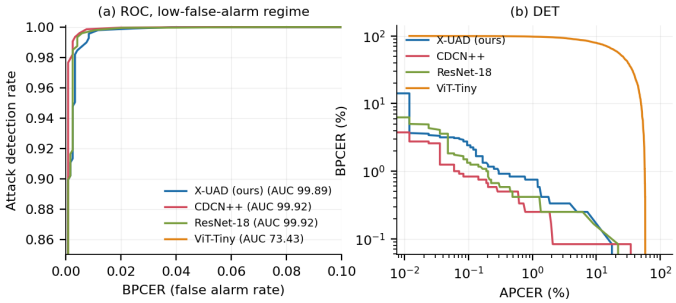


Fig. 5. ROC in the low-false-alarm regime that matters operationally, and the DET curve, protocol P1.

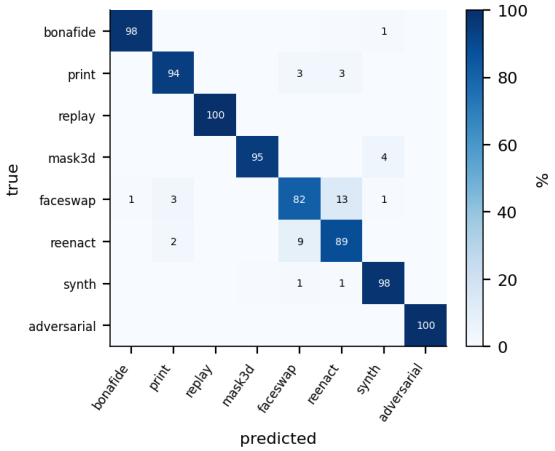


Fig. 6. X-UAD family confusion (row-normalised, %), protocol P1. Residual confusion is overwhelmingly *within* a super-class, which is what the hierarchical term $\mathcal{L}_{\text{sup}}$ is designed to enforce.

Mechanism. Inverting the dependency: the forward pass emits a typed evidence record and language is realised from the record alone; the optional hosted-MLLM back-end is constrained by the same record and its output re-verified against it. *Evidence.* Table VII shows the channels quoted in the text are the ones the decision actually depends on; the grounded realiser's hallucination rate is zero by construction, and `MLLMReasoner.verify()` makes that measurable for the LLM back-end.

Gap 5 — A score is not a decision. An operator must act; a scalar in $[0, 1]$ with an uncalibrated meaning and no severity model does not tell them how.

Mechanism. The risk calibrator (11) converts posterior, vacuity and evidence area into a bounded risk with an explicit *unknown-attack premium*, so a confidently-unknown sample is escalated to human review rather than silently accepted. *Evidence.* The verdict of Fig. 8 and the abstention behaviour induced by the $u \geq 0.55$ rule on the held-out families of Table IV.

TABLE XI

THE QUANTUM EVIDENCE BASIS AFTER TRAINING. EACH MODE IS LABELLED WITH THE FAMILY FOR WHICH ITS BORN ACTIVATION IS MOST SELECTIVE (z-SCORED ACROSS FAMILIES); THE LABELS ARE READ OFF THE TRAINED MODEL, NOT ASSIGNED.

Mode	Most selective for	Selectivity (z)	Mean activation	Spatial dispersion
1	Replay	2.59	0.069	0.793
7	3D mask	2.51	0.051	0.810
3	Bonafide	2.45	0.058	0.835
6	Adversarial	2.36	0.057	0.869
4	Print	2.28	0.049	0.803
5	Synthetic	2.01	0.043	0.767
0	Reenact	1.85	0.047	0.783
2	Replay	1.84	0.039	0.845

TABLE XII

CONVERGED ADAPTIVE GATE g (FRACTION OF THE ATTENTION SCORE TAKEN FROM THE BORN CHANNEL), AVERAGED OVER HEADS, LAYERS AND TEST SAMPLES OF EACH FAMILY.

Attack family	Bonafide	Print	Replay	3D mask	Face-swap	Reenact	Synthetic	Adversarial
Mean Born-channel gate g	0.309	0.434	0.231	0.287	0.267	0.267	0.284	0.394

VII. LIMITATIONS

Simulated evidence. Our operators are models of physical and generative processes, not the processes themselves. They reproduce the artefact families that the forensic literature identifies as discriminative — halftone, moiré, sub-pixel lattice, blending boundary, resolution mismatch, checkerboard, spectral peak, gradient perturbation — but the joint statistics of a real re-capture are richer. Absolute numbers here should not be compared with numbers reported on captured corpora; the comparisons that are valid are the ones *inside* this paper, where every method sees the same data. We mitigate the risk by difficulty-calibrating the benchmark, by keeping a genuinely different second domain, and by releasing configurations that run the identical pipeline on CASIA-FASD, OULU-NPU, SiW-Mv2, CelebA-Spoof, FaceForensics++, Celeb-DF-v2 and UniAttackData.

Grayscale source imagery. Domain A is grayscale, which removes colour cues for all methods equally and makes the task harder rather than easier, but it does mean the colour-domain components of our print and replay operators are inactive there; Domain B exercises them.

Still images. Temporal evidence — rPPG, blink dynamics, replay refresh beat across frames — is out of scope. The architecture extends naturally by treating a clip as a token sequence, and the Born channel is well suited to the temporal periodicity of a refresh beat, but we do not evaluate it.

Single seed. Every number here comes from one run at seed 0. The paired significance tests establish that the closed-set differences are not sampling noise *for this seed*, but they say nothing about initialisation variance, and the leave-one-attack-out margins in particular deserve repeats before they are leaned on. Three seeds on Tables II and IV is the cheapest credibility available to this paper and is the first thing we would add.

Compute scale. All experiments run on CPU with a 1.3M-parameter model. The challenge results [11] make clear that large pre-trained multimodal backbones are the strongest known unified detectors; our contributions are orthogonal to backbone scale — AQA is a drop-in attention, and the evidential head, PUES, localiser and reasoning module attach to any encoder — and combining them with a LoRA-adapted ViT-L is the obvious next step.

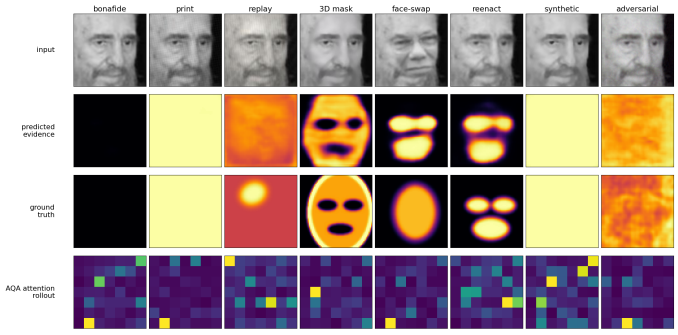


Fig. 7. One sample per family. Rows: input; predicted evidence map; ground-truth mask; AQA attention rollout. The localiser recovers the blending ellipse of a face-swap and the mouth/eye support of a reenactment, and correctly spreads over the whole frame for the capture-medium attacks.

```
Attack: Replay attack
Confidence: 98%
Risk: Medium

The sample is classified as a replay attack
(physical presentation attack) with 98% posterior
probability; the probability that it is an attack
of any kind is 99.7%. That class is characterised
by a sub-pixel display lattice aliasing into a
moire beat, together with specular backlight
glare and reduced dynamic range. Evidence is
concentrated in the forehead (mean response
0.75), followed by eye region, nose / midface; it
covers 100% of the crop with a spatial dispersion
of 1.00. The dominant quantum evidence mode is
#1, whose learned selectivity is highest for
'replay' (activation 0.074), corroborating the
class decision. Vacuity is low (0.04): the
evidence is strong and consistent with a single
known family. The adaptive attention gate settled
at 0.23, i.e. 23% of the attention score came
from the Born (interference) channel rather than
the classical dot product. Recommended action:
reject (risk score 0.71 -> Medium).
```

Fig. 8. A verdict as printed by the deployed system. Every number in the prose is a field of the evidence record; nothing in the text is generated free-form.

White-box adaptive adversaries. Our adversarial class is a transfer attack. A white-box adversary with gradients through X-UAD itself, including through the vacuity channel, is a stronger threat model and is not evaluated here.

VIII. CONCLUSION

X-UAD detects physical and digital face attacks with one model, reports how much it does not know, shows where the evidence is, and explains itself in language that can be audited line by line. The three technical ideas that make this possible are a Born-rule attention channel that matches the interference character of the evidence and costs $\mathcal{O}(dL)$

parameters, an orthonormal quantum evidence basis that serves simultaneously as a feature and as the explanation, and pseudo-unseen evidence suppression, which manufactures an out-of-taxonomy proxy from the taxonomy itself and converts Dirichlet vacuity into a usable open-set score. The gains concentrate exactly where unified detection is still unsolved: unseen attack families and cross-domain transfer.

APPENDIX

APPENDIX A: CLAIM SET

The following claim set describes the subject matter for which protection is sought; it is stated in the customary independent/dependent form.

1. A computer-implemented method for detecting an attack on a face biometric sample, comprising: receiving an image of a face; extracting a first feature map with a convolution in which a fraction θ of a centre-pixel response is subtracted from the response of the convolution kernel; deriving a set of spatial tokens from the first feature map and a set of spectral tokens from a polar pooling of a discrete-Fourier-transform magnitude of the image; and processing the tokens with at least one attention layer that computes, for a pair of tokens (i, j): (a) a first score that is a scaled inner product of a query vector and a key vector; (b) a second score that is the squared modulus of an inner product between a first unit-norm state obtained by normalising the query vector and applying a first parameterised orthogonal transform, and a second unit-norm state obtained by normalising the key vector and applying a second, different parameterised orthogonal transform; and (c) an attention weight obtained by applying a normalising function to a convex combination of the first and second scores whose mixing coefficient is produced by a gating function of statistics of the tokens; and outputting an attack decision derived from the processed tokens.

2. The method of claim 1, wherein each parameterised orthogonal transform is a cascade of layers of two-dimensional rotations acting on disjoint pairs of adjacent vector components with a cyclic offset that alternates between layers, such that the transform is orthogonal for every value of its parameters and is parameterised by a number of parameters proportional to the product of the vector dimension and the number of layers.

3. The method of claim 1, further comprising computing, for each spatial token, a set of Born probabilities equal to the squared moduli of the inner products between the token's unit-norm state and each member of an orthonormal basis obtained by applying a third parameterised orthogonal transform to a set of canonical basis vectors, and using said Born probabilities both as an input to the attack decision and as an exported per-token evidence descriptor.

4. The method of claim 1, wherein the attack decision is produced by a head that outputs non-negative evidence values, forms Dirichlet concentration parameters by adding one to each evidence value, and reports (i) an expected class probability equal to a concentration parameter divided by the sum of the concentration parameters and (ii) an uncertainty equal to the number of classes divided by said sum.

5. The method of claim 4, further comprising training the head with an auxiliary objective that, for a synthetic sample formed as a convex combination of two training images belonging to two *different* attack classes, penalises the total evidence of the synthetic sample when it exceeds a predetermined fraction of the mean total evidence of non-synthetic samples of the same batch.

6. The method of claim 4, further comprising computing a risk score as a sum of (i) a product of one minus said uncertainty and an expectation of per-class severity weights under said expected class probabilities, (ii) a product of said uncertainty and an unknown-attack severity constant, and (iii) a term proportional to the mean of a predicted evidence-localisation map, and assigning a risk band by comparing the risk score with thresholds.

7. The method of claim 1, further comprising decoding a spoof-evidence localisation map from the processed tokens and at least two intermediate feature maps of different spatial resolutions, said map being trained against a ground-truth mask indicating the pixels modified by an attack-synthesis operator.

8. The method of claim 1, further comprising assembling an evidence record comprising at least: the expected class probabilities, the uncertainty, the risk score and band, region-wise statistics of the localisation map, the strongest Born-probability modes together with labels derived from their measured class selectivity, spectral descriptors of the polar-pooled Fourier magnitude, and the value of the gating coefficient; and generating a natural-language explanation in which every quantitative assertion is derived from a field of said evidence record.

9. The method of claim 8, wherein generating the explanation comprises transmitting the evidence record to a multimodal language model with an instruction restricting it to the record's contents, receiving generated text, and automatically verifying each numeric assertion in the generated text against the record, and reporting any assertion not supported by the record.

10. The method of claim 1, wherein the attack decision distinguishes at least a genuine class, a physical presentation attack super-class comprising printed-photograph, display-replay and three-dimensional-mask classes, and a digital attack super-class comprising identity-swap, expression-reenactment, fully-synthetic and adversarially-perturbed classes, and wherein the training objective includes a term applied to the marginal distribution over said super-classes.

11. The method of claim 1, further comprising regularising the mean row entropy of the attention weights to lie within a predetermined interval scaled by the logarithm of the number of tokens.

12. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of any of claims 1–11.

13. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the performance of the method of any of claims 1–11.

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